PALESTINE POLYTECHNIC UNIVERSITY

College of Information Technology and Computer Engineering

Department Of Computer Engineering

Smart Shopping Cart and Automated Payment System (NEOSHOP)

Team members

Arwa Hajahjah - 231155

Belal Edoor - 201089

Momen Bhais - 221073

Supervisor

Wael Al Takrouri

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Abstract

Traditional in-store shopping is often a laborious and time-consuming process. Customers frequently encounter long queues and a lack of information about products and their suitability to their preferences. Furthermore, errors in billing can occur. Businesses also face significant challenges, including a lack of data for customer management and difficulties in managing inventory, making them prefer innovative solutions. Existing solutions are often either limited in scope or cost-effective, rendering them unsuitable for small businesses.
Our project offers a smart and integrated solution to enhance the shopping experience, making it more efficient and seamless. The advanced system integrates several artificial components, including barcode scanning, radio frequency identification (RFID), computer vision, and artificial intelligence, improving the overall shopping experience from product selection to payment.
The system includes a seamless shopping cart payment solution using RFID technology, as well as an automated cash register capable of accurately counting change, eliminating the need for traditional payment terminals. The system also provides essential insights into product information and allows smart leaders to receive customer preferences and health data, helping them make better decisions.
Furthermore, the system utilizes computer vision technology to monitor product movement within the store and detect unscanned items. It sends alerts to the user and can take actions such as closing the checkout window upon request. The system also provides control over the company's technical dashboard, enabling it to analyze sales data and understand customers.
The system is characterized by its low cost and scalability, making it suitable for complex environments while delivering high performance at a low cost.

Keywords: Smart Shopping Cart, Automated Payment System, Cashierless Retail, Self-Checkout, Smart Cart, Retail Automation, Automated Cash Register, Anti-Theft Detection, Product Recommendation, Store Analytics Dashboard.

الملخص

تُعد عملية التسوق التقليدية في المتاجر من العمليات التي تستهلك وقتًا وجهدًا كبيرين، حيث يعاني العملاء من الازدحام وطوابير الانتظار الطويلة وعدم الإلمام بالمعلومات الكافية حول المنتجات وهل هي مناسبة لتفضيلاتهم أم لا، بالإضافة إلى احتمالية حدوث أخطاء في الفوترة. كما يواجه أصحاب المتاجر تحديات تتعلق بنقص البيانات حول سلوك العملاء وصعوبة إدارة المخزون، فضلًا عن الخسائر الناتجة عن السرقات. الحلول الحالية غالبًا ما تكون إما محدودة الإمكانيات أو مرتفعة التكلفة، مما يجعلها غير مناسبة لمعظم المتاجر، خاصة الصغيرة والمتوسطة.

يقترح مشروعنا حلاً ذكيًا ومتكاملًا لتحسين تجربة التسوق وجعلها أكثر كفاءة وسلاسة. يعتمد النظام على دمج عدة تقنيات حديثة تشمل قراءة الباركود، وتقنية تحديد الهوية بموجات الراديو، والرؤية الحاسوبية، والذكاء الاصطناعي، لتوفير تجربة تسوق متكاملة تبدأ من اختيار المنتجات وحتى إتمام عملية الدفع.

يتضمن النظام نظام دفع مدمج يدعم الدفع عبر تحديد هوية السلة بموجات الراديو الخاصة بها بالإضافة إلى نظام دفع نقدي آلي قادر على التحقق من العملات وإرجاع الباقي بدقة، مما يلغي الحاجة إلى نقاط الدفع التقليدية. كما يتيح النظام للعملاء عرض معلومات المنتجات بشكل فوري، وتلقي توصيات ذكية بناءً على تفضيلاتهم وبياناتهم الصحية، مما يساعدهم على اتخاذ قرارات شراء أفضل.

بالإضافة إلى ذلك، يعتمد النظام على تقنيات الرؤية الحاسوبية لمراقبة حركة المنتجات داخل العربة والكشف عن العناصر غير الممسوحة، حيث يقوم بإرسال تنبيهات للمستخدم واتخاذ إجراءات تلقائية مثل إيقاف العربة عند الضرورة.

كما يوفر النظام لوحة تحكم متقدمة لأصحاب المتاجر تتيح تحليل بيانات المبيعات وسلوك العملاء وإدارة المخزون بكفاءة.

يتميز النظام بكونه منخفض التكلفة نسبيًا وقابلًا للتوسع، مما يجعله مناسبًا للتطبيق في بيئات المتاجر المحلية، مع تحقيق توازن بين الأداء العالي والتكلفة العملية.

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First, we express our deepest gratitude to God for the strength, patience, and ability He bestowed upon us to complete this impactful project.

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Chapter 1

Introduction

1.1 Preface

The NEOSHOP project is an integrated smart shopping system designed to improve the overall Shopping experience for both customers and store owners. Rather than focusing on a single function, the system provides a complete solution that supports different stages of the shopping process. This integration enables faster, more efficient, and more accurate shopping and payment operations.

1.2 Problem Statement

Despite recent technological advancements, the shopping experience in traditional supermarkets is often inefficient and unsatisfying for both customers and retailers. Shoppers suffer long wait times at checkout counters, especially during peak hours, sometimes spending more time waiting than actually shopping. Furthermore, manual checkout processes are prone to errors, such as scanning products multiple times, causing frustration for both customers and retailers. Another common problem is unscanned items in shopping carts, which can lead to unpaid items leaving the store and contributing to losses. Moreover, most stores lack systems to help customers make informed purchasing decisions, particularly for individuals with specific health conditions like diabetes or food allergies, forcing them to manually check product labels—a time-consuming and often overlooked task.

On the other hand, retailers face significant operational challenges, including a lack of detailed data on customer behavior, purchasing patterns, and product demand at different times. This limitation makes it difficult to optimize inventory management and product display strategies. Shoplifting also poses a significant financial concern, as even minor thefts accumulate over time in the absence of effective detection systems. Furthermore, traditional checkout processes are slow and labor-intensive, requiring a large number of employees, which increases operating costs without improving the overall customer experience.

1.3 The Importance of This Project

This system has the potential to make a real difference. The problems mentioned earlier are not simple, and solving most of them would have a significant impact. For customers, the system will save time and make the shopping experience easier, while the AI-powered recommendation feature helps in selecting the right products with minimal time and effort. For store owners, real-time data on customer behavior is invaluable, allowing them to manage their stores more efficiently and make better business decisions, while theft detection directly contributes to reducing financial losses. What distinguishes this project is our commitment to addressing these challenges without requiring expensive infrastructure, especially in our local environment, and the NEOSHOP project aims to provide a practical and affordable alternative.

1.3.1 Objectives

- **Develop a smart cart with real-time product scanning** – enabling automatic product recognition using barcode scanning and immediate display of each product's details and total cost, enhancing transparency and reducing checkout time.
- **Implement cart monitoring and item verification** – to continuously track the products within it and detect unregistered items, minimizing billing errors and preventing unpaid products from leaving the store.
- **Enable automated cart control and alert system** – that automatically stops the cart in case of rule violations, sending immediate notifications to the customer, store owner, and security personnel.
- **Provide a seamless checkout experience** – by eliminating the need to re-scan products at checkout, reducing waiting times and improving overall shopping efficiency.
- **Ensure reliable inter-system communication** – by using a message broker to handle asynchronous data exchange between system components, especially during payment and invoice processing.
- **Support automated cash payment processing** – through an intelligent system capable of accurately verifying banknotes and coins and providing change, ensuring a reliable and user-friendly payment process.
- **Develop a management and analytics platform** – to help store owners track sales, manage inventory, and generate invoices, while providing valuable data to support decision-making.

- **Implement a personalized recommendation system** – to suggest products based on user preferences and shopping behavior, enhancing the overall customer experience.
- **Enable in-store product navigation assistance** – to help customers easily locate products inside the store using smart guidance technologies.

Chapter 2

Literature Review and Theoretical Background

2.1 Preface

This chapter consists of two sections: the first reviews current smart shopping systems and their methodologies, while the second explains the key technologies used in the NEOSHOP system and the reasons for their selection, providing a foundation for the system's design.

2.2 Literature Review

This section reviews existing smart shopping systems developed both internationally and locally. It aims to analyze their methodologies and identify their strengths and weaknesses. This analysis helps clarify the gap that the NEOSHOP system is designed to address.

2.2.1 computer vision

Amazon Go [1] stores use computer vision technology as a primary mechanism for product identification, continuously tracking items received or returned by shoppers without any manual scanning. The smart shopping system [2] The system also uses computer vision technology to identify currency: it illuminates the input banknotes with an ultraviolet LED and then captures an image of them using a webcam. The OpenCV library then applies a threshold to isolate the UV-reactive watermark area, and Tesseract OCR software reads the currency value from the glowing watermark. If no text is found, the banknote is rejected as counterfeit. Coins are identified by measuring their diameter using the HoughCircles algorithm and verifying their characteristics by comparing them to reference images using FLANN matching. In our NEOSHOP project, we use computer vision to quickly detect shoplifting attempts by tracking the movements of a customer's hand and the product they are holding to ensure it is placed correctly in the shopping cart.

2.2.2 RFID

Smart shopping system [2] Radio Frequency Identification (RFID) technology is used to identify products and process payments, automating the entire checkout process. However, this system requires labeling each product in the store, increasing operational costs. Newshop, on the other hand, uses RFID technology to identify the cart at checkout, thus avoiding the cost of labeling all products while maintaining automated payment capabilities.

2.2.3 Barcode Scanning

The smart shopping cart project [3] relies on barcode scanning for product identification, making it cost-effective by leveraging existing infrastructure. NEOSHOP adopts the same approach for product identification, retaining the cost advantage while extending it with AI-powered features and automated payment.

2.2.4 Artificial Intelligence (Analysis and Recommendations)

None of the reviewed systems, including Amazon Go [1], the smart shopping system [2], and the smart shopping cart project [3], provide personalized product recommendations or shopper analytics accessible to end users. Furthermore, these systems do not expose AI-driven features directly to shoppers or store owners.

NEOSHOP addresses this limitation by integrating an AI-powered recommendation engine together with a store owner-facing analytics dashboard. These features allow the system to suggest products based on shopping behavior, provide insights into customer preferences, and support better business decisions. As a result, the shopping cart becomes an intelligent assistant rather than a passive container.

2.3 Basic Theoretical Concepts

This section explains the main technologies used in the NEOSHOP system and the reasons for their selection. This selection is based on effectiveness, cost, efficiency, and practical feasibility.

2.3.1 barcode scanning technology

The system uses barcode scanning technology to identify products [4]. This technology was chosen for its widespread availability across all products. The system captures and decodes the information stored in the barcode using a light source, a sensor, and a decoder.

2.3.2 RFID technology

RFID technology [5] is used to store and transmit invoice data during the payment process . This method was chosen because it enables fast wireless communication via radio waves, reducing transaction time. It also improves system efficiency by allowing invoice data to be transmitted without the need for re-scanning.

2.3.3 Computer Vision Technologies

Computer vision technologies are used to enhance security and automate monitoring. The system uses YOLOv8nano [6] for fast response object detection and tracking, enabling continuous monitoring of products held by customers inside the store. The model detects and follows product movement to identify suspicious behaviors such as concealed items, unauthorized product removal, prolonged unusual handling, or attempts to leave designated areas without completing payment. Using OpenCV [7] , the system processes live video streams from surveillance cameras to improve detection accuracy and analyze customer-product interactions in real time. The monitoring module helps detect potential theft activities and suspicious actions, allowing the system to trigger alerts and support smart retail security operations automatically.

2.4 Summary

The selected technologies provide a balance between performance, cost, and practicality. Unlike existing solutions that are either complex or limited, NEOSHOP combines efficient algorithms and scalable approaches to deliver a comprehensive and realistic smart shopping system.

Chapter 3

Requirements and System Analysis

3.1 Preface

This chapter presents the system requirements and analysis for the NEOSHOP system. It identifies functional and non-functional requirements, explains how users interact with the system through use cases, and provides a conceptual overview of the system and its presence in the environment.

3.2 System Requirements

3.2.1 Functional Requirements

The NEOSHOP system provides many key functions for customers and shop owners.

- The system shall allow users to scan products using a barcode scanner integrated into the cart.
- The system shall display product details such as name, price, and ingredients.
- The system shall calculate and update the total bill dynamically.
- The system shall alert users if a product is not suitable according to their health conditions.
- The system shall provide recommendations for AI-based products.
- The system shall display a store map to help users locate products.
- The system shall support automated cash payment with return of change.
- The system shall provide a dashboard to monitor sales and invoices.

- The system must analyze product consumption and sales flow.
- The system shall generate reports on customer behavior and product trends.

3.2.2 Non-Functional Requirements

- **Performance:**There should be a quick entry time for scanning and detecting products, ensuring a smooth and immediate shopping experience without any noticeable delay.
- **Ease of Use:** The system should be intuitive, simple, and easy to use for all users, including customers and store staff, with a clear interface and minimal complexity of interaction.
- **Reliability:** The system should operate consistently without errors, especially during invoice creation, product detection, and payment processing, ensuring accurate and reliable results.
- **Scalability:** The system should support increasing numbers of smart carts and payment units without performance degradation, allowing easy expansion as the store grows.
- **Cost-Effectiveness:** The system must operate without errors, especially during invoice creation, product detection, and exclusion, ensuring accurate results.

Security Requirements

- The system should detect suspicious movements and attempted thefts using computer vision technology.
- The system shall alert the user when suspicious behavior is detected.
- The system will activate the vehicle's brakes if theft is confirmed and notify store owners and security personnel in case of theft.
- The system shall ensure secure transmission of data between system components using encrypted communication protocols.
- The system shall protect sensitive data (such as user preferences and transactions) using secure storage mechanisms.
- The system must verify the authenticity of the entered banknotes to detect counterfeit currency.
- The system shall ensure that payment transactions are completed securely and accurately without manipulation.

3.3 System Interaction Analysis

3.3.1 Main Actors

- Customer
- Store Owner
- Security staff

3.3.2 System interactions analysis

Use Case Name: Smart Shopping Process

Primary Actor: Customer

Secondary Actors: Security Staff, Store Owner

Goal: To enable the customer to shop, automatically detect theft, complete the payment process, and provide the store owner with quick analytics, all through the NEOSHOP smart shopping cart system.

3.3.3 Preconditions:

- The customer was registered and his identity was verified in the system.
- The smart cart is powered on and connected to the backend.
- Products in the warehouse are recorded in the database using valid numerical codes.
- The payment unit is loaded with sufficient coins for change.

3.3.4 Main Flow

- The customer scans a product using the barcode scanner on the smart cart.
- The system displays the product details and price, then updates the bill total in real time.
- The system analyzes the product against customer health preferences and suggests AI-based product recommendations if a better alternative exists and location on the in-store map shown on the cart screen.
- The customer adds the product to the cart.
- The customer can remove any product from the shopping cart at any time by pressing the ”Remove” button and scanning it.

- The system sends warning notifications to the user via the screen if suspicious activity is detected.
- When the customer finishes shopping, they press the "Checkout" button. The invoice is stored in the messaging system, ensuring its delivery to the payment system even in case of malfunctions or errors. The customer then proceeds to the payment terminal. The system identifies the cart via RFID technology, and the invoice information is transmitted backend via the broker.
- The customer receives their change from the payment system.
- After the invoice is generated, the system transmits all transaction data to the store owner's dashboard.
- The store owner views sales performance reports generated by the analytics engine.
- The store owner monitors purchasing trends using Apriori association and linear regression models.
- The store owner manages inventory stock data.
- The store owner and security staff can view all suspicious activity notifications and event logs at any time.

3.3.5 Alternative Flows

- If the barcode is unreadable, the system displays an error message and prompts the customer to try again.
- If a counterfeit banknote is detected, the system rejects it and alerts the customer to insert a valid note.
- If the customer removes a product before checkout, the system updates the invoice and RFID tag accordingly.
- If the RFID read fails at checkout, the customer is prompted to retry before payment proceeds.

3.3.6 Postconditions

- The completed invoice and payment record are stored in the database (MySQL, ACID-compliant).
- The shopping session data, cart log, and behavioral analytics are transmitted to the store owner's dashboard for review.
- Stock quantities in the Product table are updated to reflect purchased items.
- Any security events (theft alerts, brake activations) are logged in the SecurityEvent table and remain available for audit.

Figure 3.1 shows the interaction between the system and its users.

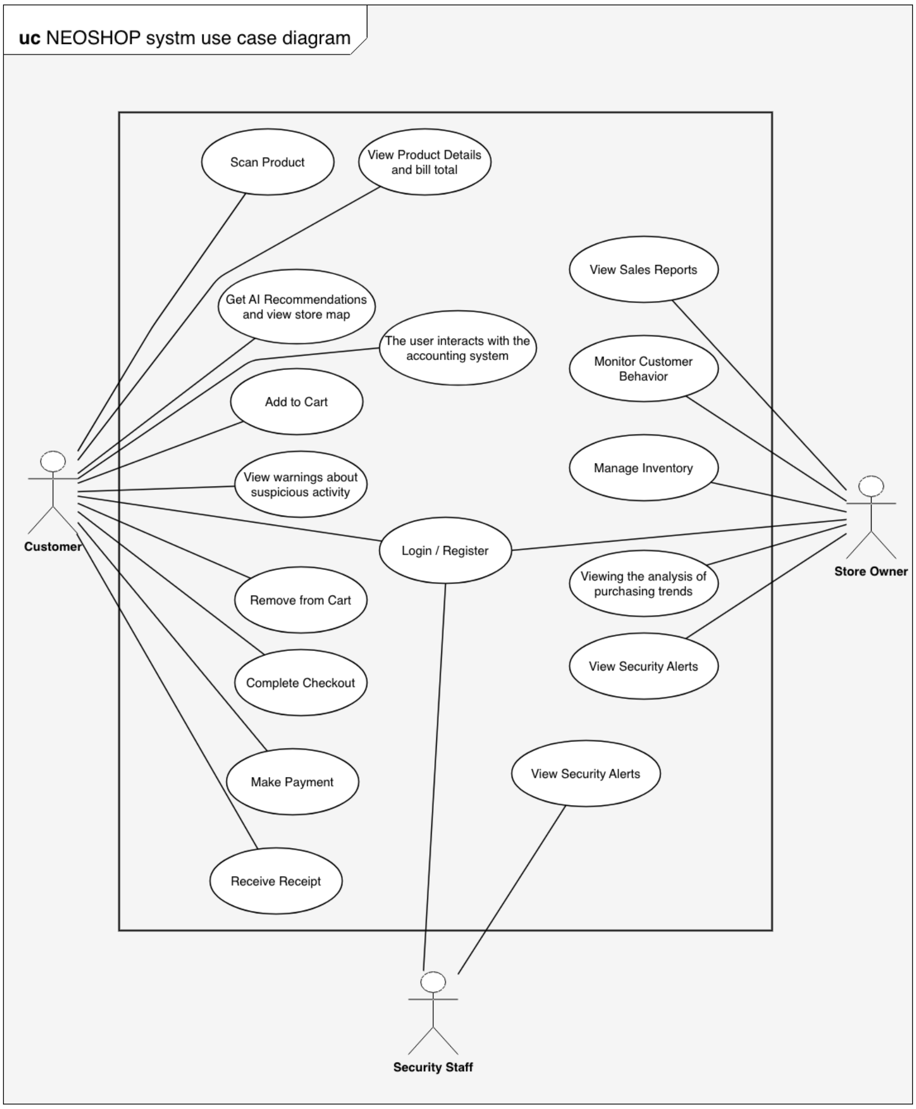


Figure 3.1: Use Case Diagram of NEOSHOP System

3.3.7 Conceptual System View

3.3.7.1 Overview

To better understand the overall structure and data flow of the proposed NEOSHOP system, a general diagram is provided below (as shown in Figure 3.2). This diagram offers a simplified overview of the main system components and illustrates how they are interconnected to execute the shopping and payment process efficiently.

3.3.7.2 How the System Works

The system's first core component is a smart shopping cart equipped with a barcode scanner, a small display screen, and a built-in camera. The customer begins shopping by scanning each product with the scanner. Once a product is scanned, the system retrieves its data and displays it on the cart screen, including the name, price, and ingredients, and instantly updates the total bill.

Simultaneously, an AI module analyzes the scanned product based on the customer's preferences and health status. If a better or more suitable alternative is available, the system suggests it and displays its location on a store map shown on the cart screen, making it easier for the customer to find it themselves.

As the customer places products in the cart, the built-in camera monitors all hand movements and product interactions using computer vision algorithms. If the system detects a product being placed in the cart without first scanning it, it immediately alerts the customer on the screen. If the alert is ignored, the shopping cart's brakes are automatically activated to prevent it from moving, and an immediate notification is sent to both security personnel and the store owner.

When the customer finishes shopping, they proceed to the checkout area. An RFID reader located at the checkout area reads the entire invoice via the RFID chip embedded in the shopping cart.

The invoice data is then transmitted through a message intermediary that acts as a bridge between the shopping cart and the payment system, ensuring synchronization and system reliability.

The customer then pays using an automated cash register, which accepts and verifies banknotes and coins, automatically calculates the change, and dispenses the correct amount.

After payment, all transaction data and purchase behavior data are sent to the store owner's dashboard. This dashboard provides detailed reports on sales performance, inventory movement, product popularity, and purchasing trends, enabling data-driven decisions regarding product display, inventory management, and marketing strategies.

Figure 3.2 shows the interaction between the system and its users. illustrates the conceptual architecture of the NEOSHOP system,

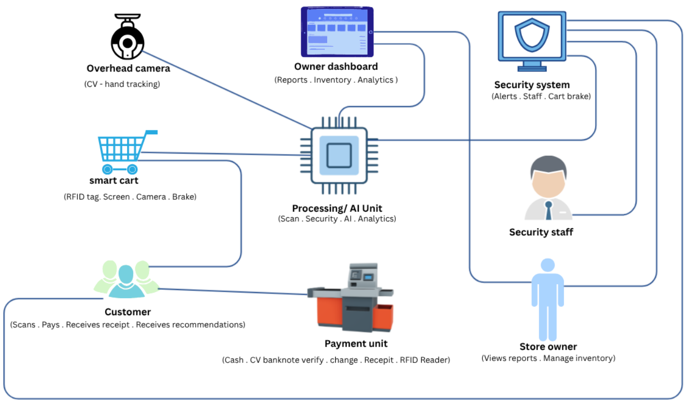


Figure 3.2: Conceptual System View of NEOSHOP

3.4 Database Analysis

3.4.1 Overview

The database analysis phase identifies the core data entities required by the NEOSHOP system, their attributes, and the relationships between them.

Table 3.1: Database Entities, Attributes, and Relationships

Entity	Attributes	Relationships
User	UserID (PK), Name, Email, Password, Phone, is_active	One user performs 1:N ShoppingSessions
Customer	UserID (PK), age, gender, allergies,	Inherits from User (1:1)
Owner	UserID (PK), business_name, license_number	Inherits from User (1:1)
Security	UserID (PK), badge_number, shift_schedule, clearance_level	Inherits from User (1:1)
ShoppingCart	CartID (PK), Status	One cart is used in 1:N ShoppingSessions
Product	Barcode (PK), Name_ar, Price, StockQuantity, Category, Description, Ingredients, Allergens, Section	One product appears in 1:N CartItems

Continued on next page

Table 3.1 – continued

Entity	Attributes	Relationships
ShoppingSession	SessionID (PK), CartID (FK), UserID (FK), StartTime, EndTime, Status	Belongs to N:1 User; Belongs to N:1 ShoppingCart; Generates 1:1 Transaction; May trigger 1:N Alerts
Transaction	TransactionID (PK), SessionID (FK), Method, total_amount, discount, amount_inserted, change_returned, status, created_at, completed_at	Generated by 1:1 ShoppingSession; Associated with 1:1 Invoice
Invoice	InvoiceID (PK), SessionID (FK), Cart_ID (FK), Total_Amount	Associated with 1:1 Transaction
Alert	AlertID (PK), SessionID (FK), UserID (FK), AlertType, description, BrakeActivated, resolved, created_at, resolved_at	Triggered by N:1 ShoppingSession; Created by N:1 User

3.4.2 Business Rules and Constraints

The following constraints govern data integrity across the system:

- A shopping session cannot be created without a valid authenticated user account.
- A transaction record cannot be created unless a completed shopping session exists.
- An `RFIDInvoice` must always be linked to an existing transaction and cannot exist independently.
- An `Alert` must always reference an existing shopping session.
- Every product must have a unique barcode and every user must have a unique email address.
- Stock quantity in the `Product` table is updated after each completed transaction.

3.5 Summary

This chapter outlines and analyzes the requirements of the NEOSHOP system. It addresses the functional requirements for customers, security, and store owners, as well as the non-functional requirements related to performance, reliability, and scalability. The use case analysis describes the entire smart shopping process, while the conceptual presentation illustrates how the smart shopping cart system and the RFID payment system interact during the purchase process, with both systems feeding into a shared backend that supports the store owner's dashboard and analytics.

Chapter 4

System Design

4.1 Preface

This chapter presents and discusses the overall system design of the NEOSHOP Smart Shopping Cart and Automated Payment System, illustrating how its various hardware and software components are integrated to operate as a unified and efficient solution. It explains the structural organization of the system and how different modules interact to achieve the desired functionality.

The chapter includes a detailed presentation of the system block diagram and schematic design, which provide a clear overview of the internal architecture and connections between components. In addition, it introduces the key algorithms used in the system, highlighting their role in enabling core functionalities such as item detection, recommendation, and payment processing.

Furthermore, a concise description of the selected hardware and software components is provided, along with the rationale behind their selection. This ensures a clear understanding of how each part contributes to the overall system performance, reliability, and scalability.

4.2 System Components and Design Alternatives

4.2.1 Hardware Components and Options

The hardware components of the NEOSHOP system were carefully evaluated based on criteria such as computational capability, power consumption, cost, ease of integration, and scalability. Table 4.1 summarizes all hardware alternatives considered, the selected component for each category, and the justification for each choice.

Table 4.1: Hardware Components: Alternatives, Selections, and Justifications

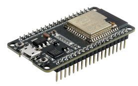
Category	Alternatives Considered	Selected Component	Reason for Selection	Image
Smart Cart Controller	Raspberry Pi 4 Model B [8], ESP32 [9], Arduino Mega [10]	Raspberry Pi 4 Model B [8]	Raspberry Pi 4 leads with a quad-core 1.8 GHz CPU, 8 GB RAM, and native AI inference, Power up to 8 watts, making it the most powerful option at 55 Dollar.	
Payment System Controller	ESP32 [9], Arduino Uno [11], Raspberry Pi 4 [8]	ESP32 [9]	The ESP32 is the ideal pulse controller, featuring fast pulse calculation, built-in Wi-Fi and Bluetooth, power consumption of no more than 0.5 watts, and a price of only $40.	
Barcode Scanner	Laser Barcode Reader [12], 2D Camera Scanner [13]	Laser Barcode Reader [4]	Fastest decode latency (<50 ms) with no host-side processing; all commercial products use 1D barcodes.	
Camera Module	USB Webcam [14], Raspberry Pi Camera v2 [15]	USB Webcam [14]	Plug-and-play UVC; no closed-source drivers or MIPI config needed; flexible across Linux platforms.	
Display	7-inch HDMI Touchscreen [16], Standard Monitor	7-inch HDMI Touchscreen [17]	High-resolution (1024×600) capacitive multi-touch, stable HDMI, seamless Raspberry Pi integration.	
Braking System	Servo-lock Mechanism [18], Electromagnetic Brake [19]	Servo-lock Mechanism [18]	Zero steady-state current draw; more efficient than electromagnetic brake (0.5–2 A continuous).	
RFID Module	RC522 (SPI) [20], PN532 (I2C) [21]	RC522 [20]	Read latency <5 ms; full-duplex SPI avoids bus conflicts; lower cost ($2–$5 vs $5–$10).	

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Table 4.1 – continued

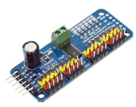
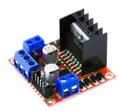
Category	Alternatives Considered	Selected Component	Reason for Selection	Image
Banknote Verification	TP74 Bill Validator [22], Computer Vision (OpenCV + UV LED + Webcam) [7, 23]	TP74 Bill Validator [22]	The TP74 banknote validation device is used to verify the authenticity of banknotes within a smart retail system. It provides reliable counterfeit detection and denomination identification, improving the accuracy and security of cash handling operations. Its integration with the ESP32 module further enhances the efficiency and reliability of automated payment processes.	
Coin Acceptor	CH-926 [24], Generic Coin Sensor	CH-926 Coin Acceptor [24]	Configurable multi-denomination recognition, ESP32-compatible pulse output, built-in counterfeit detection.	
Coin Dispenser	MG996R [25], SG90 [26], Vibration Motor [27]	MG996R + 5× SG90 + Vibration Motor [28, 29, 30]	MG996R sorts coins by denomination; SG90 servos release coins per compartment; vibration motor prevents jamming.	  
Servo Driver	PCA9685 [31], Direct GPIO PWM	PCA9685 16-Ch. Driver [32]	Offloads PWM from GPIO; I2C with 16 independent channels for precise control and future expansion.	
Motor Driver	L298N [33], L293D, DRV8833	L298N Motor Driver [33]	Dual H-bridge for DC motor control; supports 5–35 V; protects board from high motor currents.	
OLED Display	SSD1306 OLED [34], LCD 16×2	SSD1306 OLED [35]	I2C (2 signal lines); 128×64 px high-contrast low-power display; ideal for payment unit.	

Table 4.2: Software Components: Alternatives, Selections, and Justifications

Category	Alternatives Considered	Selected Component	Reason for Selection
Object Detection	YOLOv8-nano [36], MobileNet SSD [37], Faster R-CNN [38]	YOLOv8 [36]	Real-time inference at ≈30 fps on RPi 4; superior accuracy for small objects; lower cost than two-stage Faster R-CNN.
Recommendation Engine	Rule-Based + Cosine Sim. [39, 40], Collaborative Filtering [41], k-NN [42]	Rule-Based + Cosine Similarity [39, 40]	Deterministic health constraints; no historical data required; avoids cold-start problem of collaborative filtering.
Database	MySQL [43], MongoDB [44], PostgreSQL [45]	MySQL [43]	Full ACID compliance for financial data; low resource use; native SQLAlchemy integration with FastAPI backend.
Backend Framework	FastAPI, Flask, Django	FastAPI	High-performance async handling; automatic OpenAPI docs; lightweight for embedded deployment; compatible with all AI Python libraries.
Operating System	Raspberry Pi OS [46], Ubuntu Server, DietPi	Raspberry Pi OS [46]	Official OS optimized for RPi; 35 000+ precompiled packages; native GPIO control and computer vision support.
Programming Language	Python [47], C++, Java	Python [47]	Supports PyTorch, OpenCV, FastAPI, and SQLAlchemy; enables rapid development and seamless module integration.

4.3 System Architecture and Design Modeling

This section presents the NEOSHOP architecture through six complementary SysML and UML diagrams covering structural decomposition, internal wiring, process flow, interaction sequences, engineering constraints, and physical interconnections.

4.3.1 System Components and Relationships

The BDD decomposes the top-level NEOSHOP System into five primary sub-blocks: Smart Cart Unit, AI and Processing Module, Payment Unit, Store Management Dashboard, and Database Server. Each block exposes typed ports that define the information and physical flows between components, providing a complete structural overview of the system. As illustrated in Figure 4.1, the block decomposition highlights how each subsystem contributes to the overall architecture.

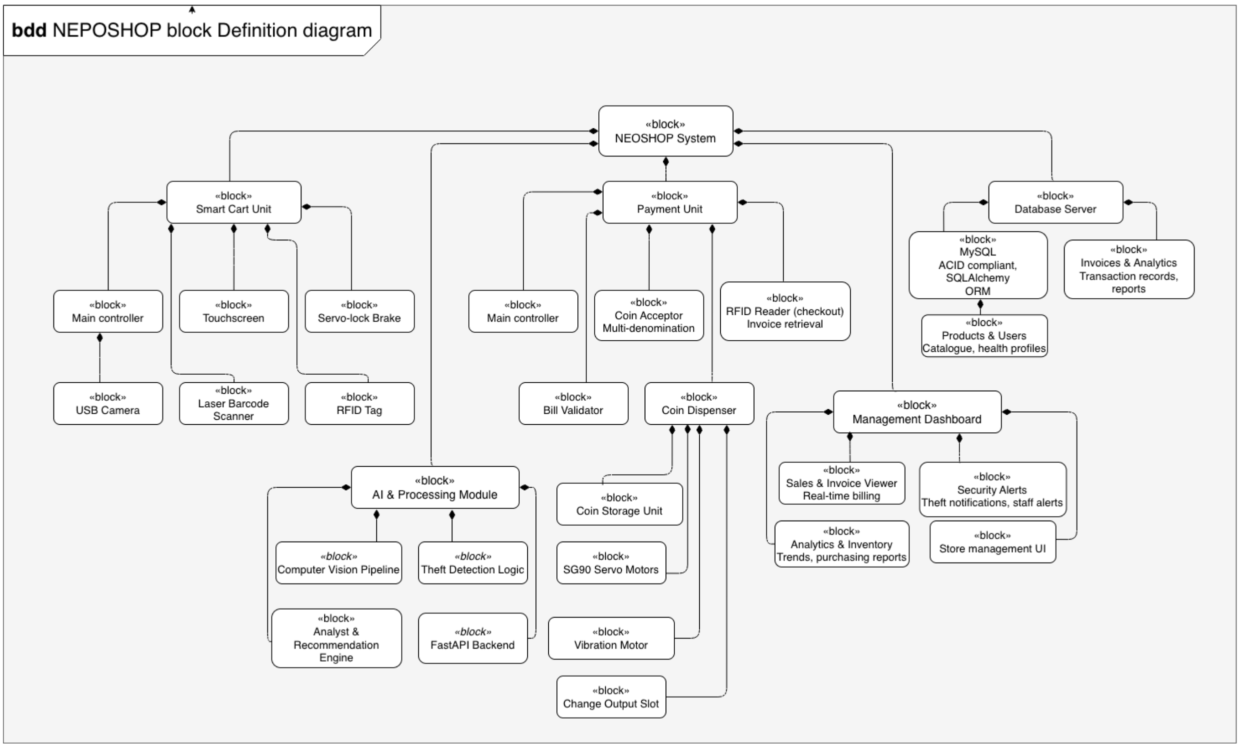


Figure 4.1: Block Definition Diagram of the NEOSHOP System

The internal structure of the AI & Processing Module is further decomposed into four sub-blocks: the Computer Vision Pipeline (YOLOv8-nano), the Theft Detection Logic (Hand Movement Monitor and Brake Controller), the Recommendation Engine (Rule-Based Filter, Cosine Similarity, Apriori Algorithm, and Linear Regression), and the FastAPI Backend (REST API and WebSocket Server). All sub-blocks follow SysML BDD compositional notation, as shown in Figure 4.2.

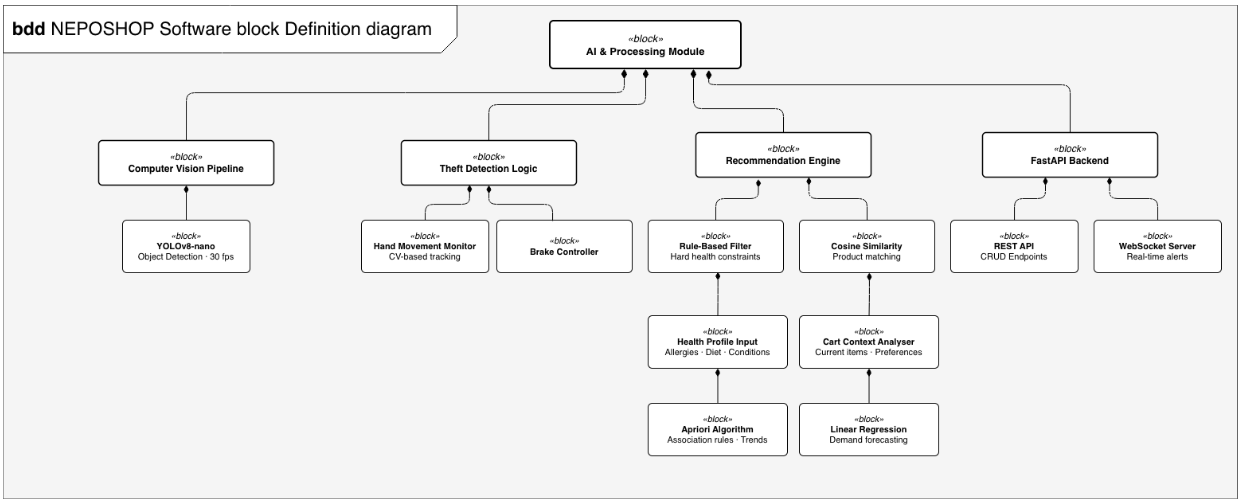


Figure 4.2: Block Definition Diagram of the NEOSHOP Software

4.3.2 Internal System Architecture and Software Design

The NEOSHOP system integrates three subsystems as shown in Figure 4.3. The Smart Cart System uses a Raspberry Pi 4 to coordinate a barcode scanner, surveillance camera, RFID unit, touchscreen display, and servo-lock brake, with all data transmitted to the backend via secure Wi-Fi using REST and WebSocket protocols.

The Software Modules employ a four-layer architecture: the AI & Processing Layer runs YOLOv8nano for theft detection and recommendations; the Application & Services Layer manages business logic through FastAPI and WebSocket with a message broker ensuring reliable invoice delivery; the Data Layer stores all transactions and events in MySQL; and the Presentation Layer provides interfaces for customers, owners, and security staff.

The Payment System operates independently with an ESP32 managing bill validators, coin acceptors, and change dispensers. The RFID reader retrieves invoices from the message broker, processes payments, displays results on an OLED screen, and sends transaction data back to the software backend, completing the integrated shopping and payment workflow.

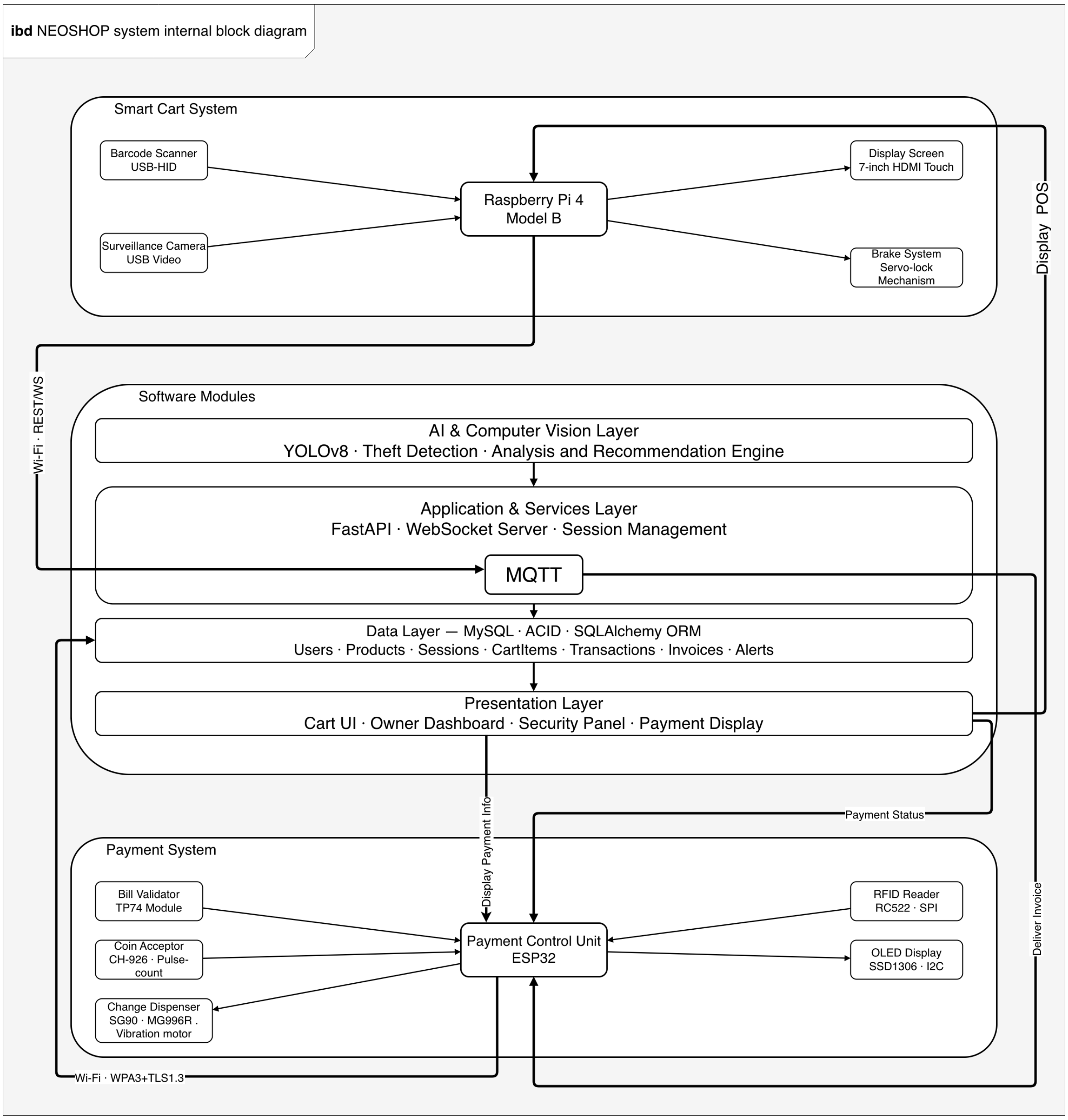


Figure 4.3: Internal Block Diagram of the NEOSHOP System

4.3.3 System Workflow

The NEOSHOP activity flow spans five concurrent swimlanes Customer, System (Raspberry Pi), Security System, Payment Unit (ESP32), and Backend System + MQTT covering the full lifecycle of a shopping session from cart activation to data synchronization.

The flow starts with account verification, where new customers register their health and allergy data while returning customers log in directly. The Raspberry Pi then loads the health profile, connects to the product database, and activates the camera. During shopping, the system retrieves product information and generates health-based recommendations while the computer vision module runs in parallel to detect suspicious activity. If an alert is ignored, the cart brake engages and security staff are notified. Once shopping is complete, the ESP32 handles payment, change dispensing, and receipt generation, after which the backend updates inventory and pushes analytics to the owner dashboard.

As shown in Figure 4.4, the parallel processing between the Raspberry Pi and Security System swimlanes is clearly visible during the scanning phase, and the branching logic at the account check and suspicious activity decision points is explicitly captured before all swimlanes converge at the session termination node.

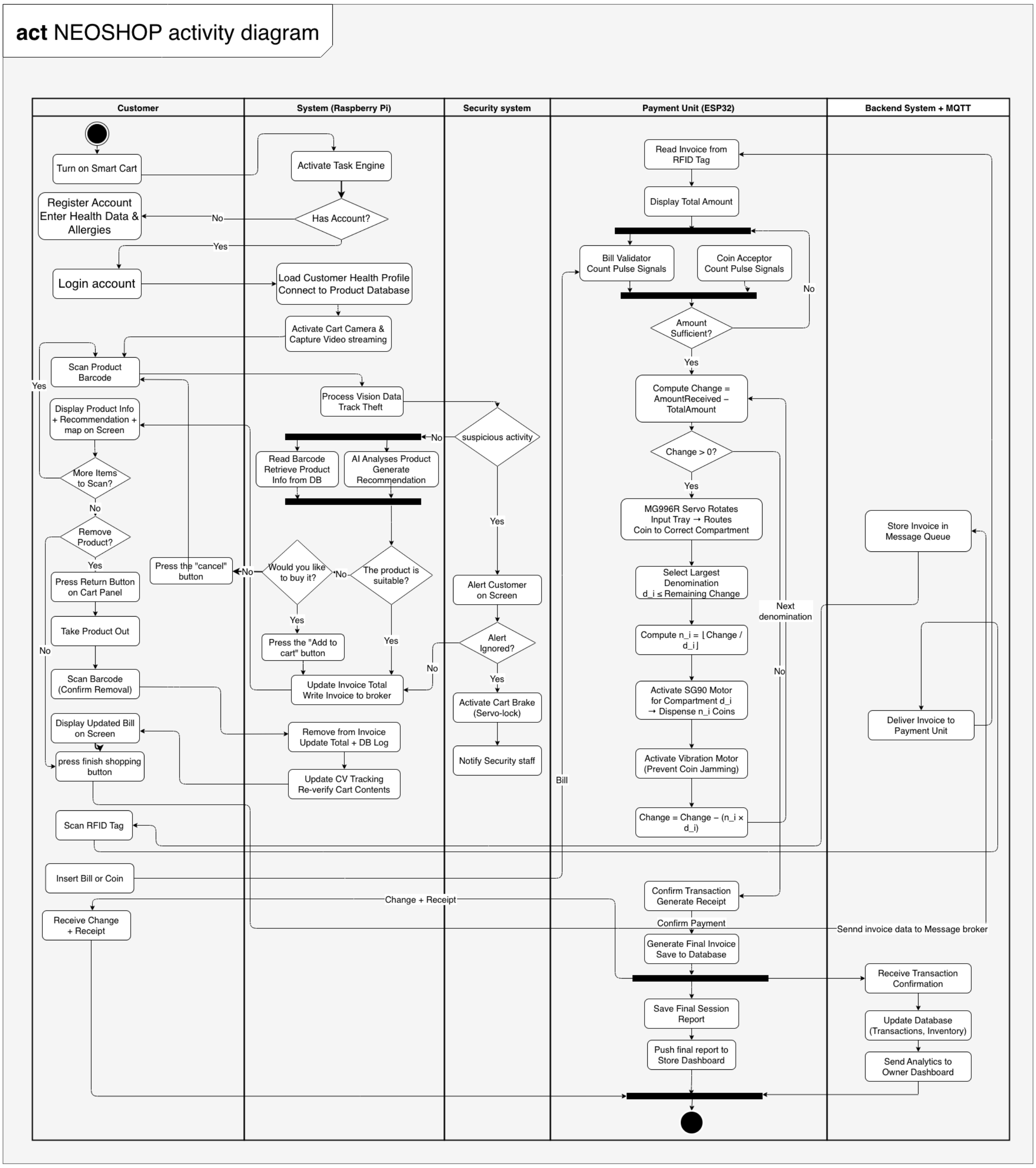


Figure 4.4: Activity Diagram of the NEOSHOP Shopping Process

4.3.4 System Interactions Flow

The cart interaction flow begins when the customer turns on the smart cart and the Raspberry Pi opens a session with the database. New customers register their health data and allergies, while existing customers log in and load their profile. The system then activates the camera and initializes the YOLOv8-nano shoplifting module.

During the shopping loop, each barcode scan triggers product retrieval, AI-based suitability analysis, and invoice and RFID tag updates, with results displayed to the customer. The Shoplifting Module monitors the cart in parallel; unscanned items trigger an alert, and if ignored, the cart brake activates and the Security System is notified. An optional removal flow allows the customer to rescan and remove items before pressing end shopping, which stores the final invoice in the Message Broker queue.

As illustrated in Figure 4.5, the nested `alt` blocks separate theft detection from normal product flow, while the `opt` block captures the removal path, and the parallel involvement of the AI Engine and Shoplifting Module confirms simultaneous recommendation and security processing.

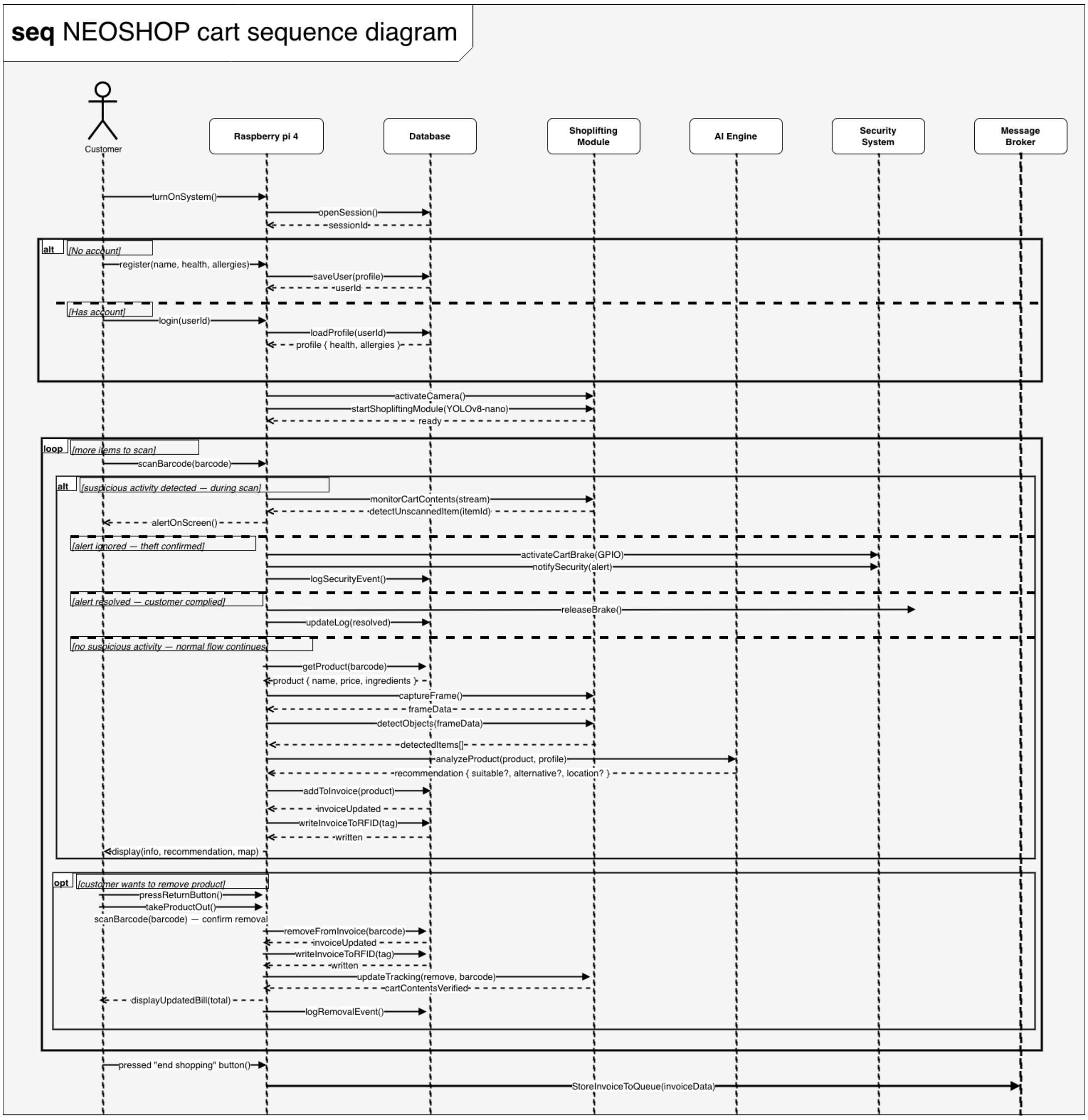


Figure 4.5: Sequence Diagram of the Cart Interaction Process Flow

Payment System

The payment process begins when the customer approaches the payment unit and scans the RFID tag embedded in the smart cart. The ESP32 retrieves the invoice data from the message broker and displays the total amount due on the OLED screen. The customer then inserts banknotes and coins into the payment unit. Banknotes are verified by the TP74 bill validator, which checks the authenticity and denomination of each note. Coins are processed via the CH-926 coin acceptor, which counts pulse signals and converts them into coin values. If a counterfeit banknote is detected, the system immediately rejects it and alerts the customer to insert a valid note. The system continues accepting cash until the total inserted amount meets or exceeds the invoice total.

Once sufficient cash is collected, the system computes the change to be returned. The total change is first calculated as:

$$C = P_{\text{inserted}} - P_{\text{total}} \tag{4.1}$$

where C is the change to be returned, P_{inserted} is the total amount inserted by the customer, and P_{total} is the invoice total. The ESP32 then applies a greedy algorithm to determine the optimal combination of coins and notes to dispense, iterating over available denominations d_i in descending order:

$$n_i = \left\lfloor \frac{C_{\text{remaining}}}{d_i} \right\rfloor, \qquad C_{\text{remaining}} = C_{\text{remaining}} - n_i \times d_i \tag{4.2}$$

where n_i is the number of coins/notes of denomination d_i to dispense, d_i is the value of denomination i sorted in descending order, and $C_{\text{remaining}}$ is the remaining change after each dispensing step. The loop terminates when $C_{\text{remaining}} = 0$. The MG996R servo motor sorts the coins by denomination while the SG90 servos release the correct number of coins per compartment, and a vibration motor prevents jamming during dispensing. Once the change is dispensed, the ESP32 confirms the transaction, generates a receipt, and transmits the complete transaction data to the backend, where the FastAPI server updates the database and pushes the session analytics to the store owner dashboard.

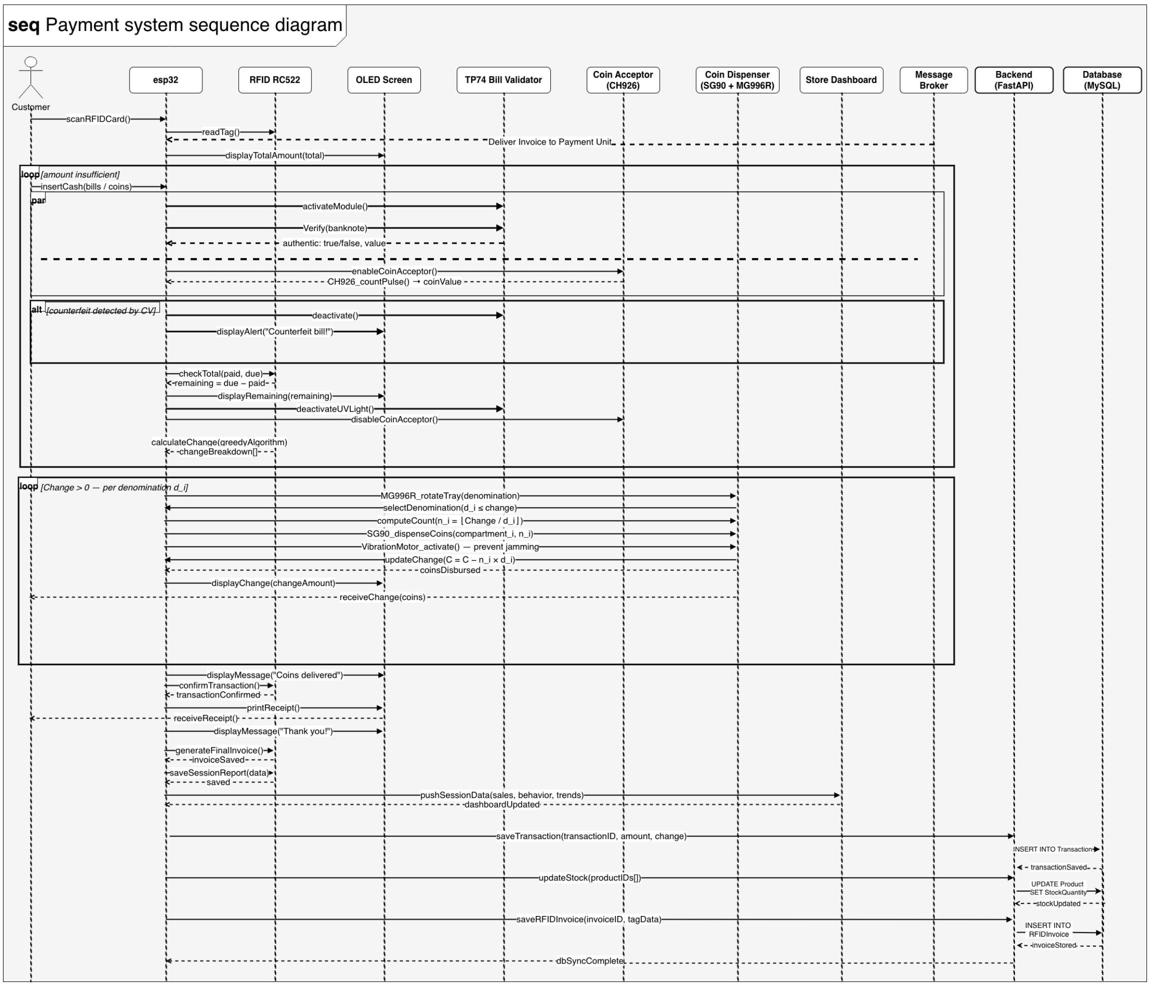


Figure 4.6: Sequence Diagram of the Payment Transaction Process Flow

4.3.5 Database Design

Database Overview:

The backend system is based on a relational database (MySQL), chosen for its reliability, ACID compliance, and efficiency in handling structured transaction data. This enables consistent storage of user information, shopping sessions, product data, transactions, and security events. The database design supports real-time operations while ensuring data integrity and scalability for the NEOSHOP system. The message broker works with the database by temporarily holding invoice data while it is being transferred between system components.

Database Schema Overview:

Table	**Description**
Users	Stores customer, store owner, and security accounts.
Products	Contains product details such as barcode, name, price, and stock.
ShoppingSession	Represents user shopping sessions and activity tracking.
CartItem	Stores items added to the cart during a session.
Transaction	Records payment details and transaction information.
Invoice	Stores invoice data transmitted via broker.
Alert	Logs theft detection events and system alerts.

Table 4.3: Database Schema Overview

Figure 4.7 shows the overall structure of the database and the relationships between its main components. The User entity represents different types of users (customer, owner, or security) and is associated with the ShoppingSession entity, which captures each user's shopping activity. During a session, products are added to the cart through the CartItem entity, which links the session to the Product entity containing details such as name, price, and stock quantity. Once the shopping process is completed, a Transaction is generated to record payment details, including the inserted amount and returned change. The system also includes an Alert entity triggered by a shopping session in case of suspicious behaviour, such as theft detection. Additionally, an RFIDInvoice entity is created and linked to the transaction, storing RFID-related data and the invoice generation time. Overall, the ERD demonstrates a well-structured and integrated system that combines user management, shopping operations, payment processing, and security monitoring.

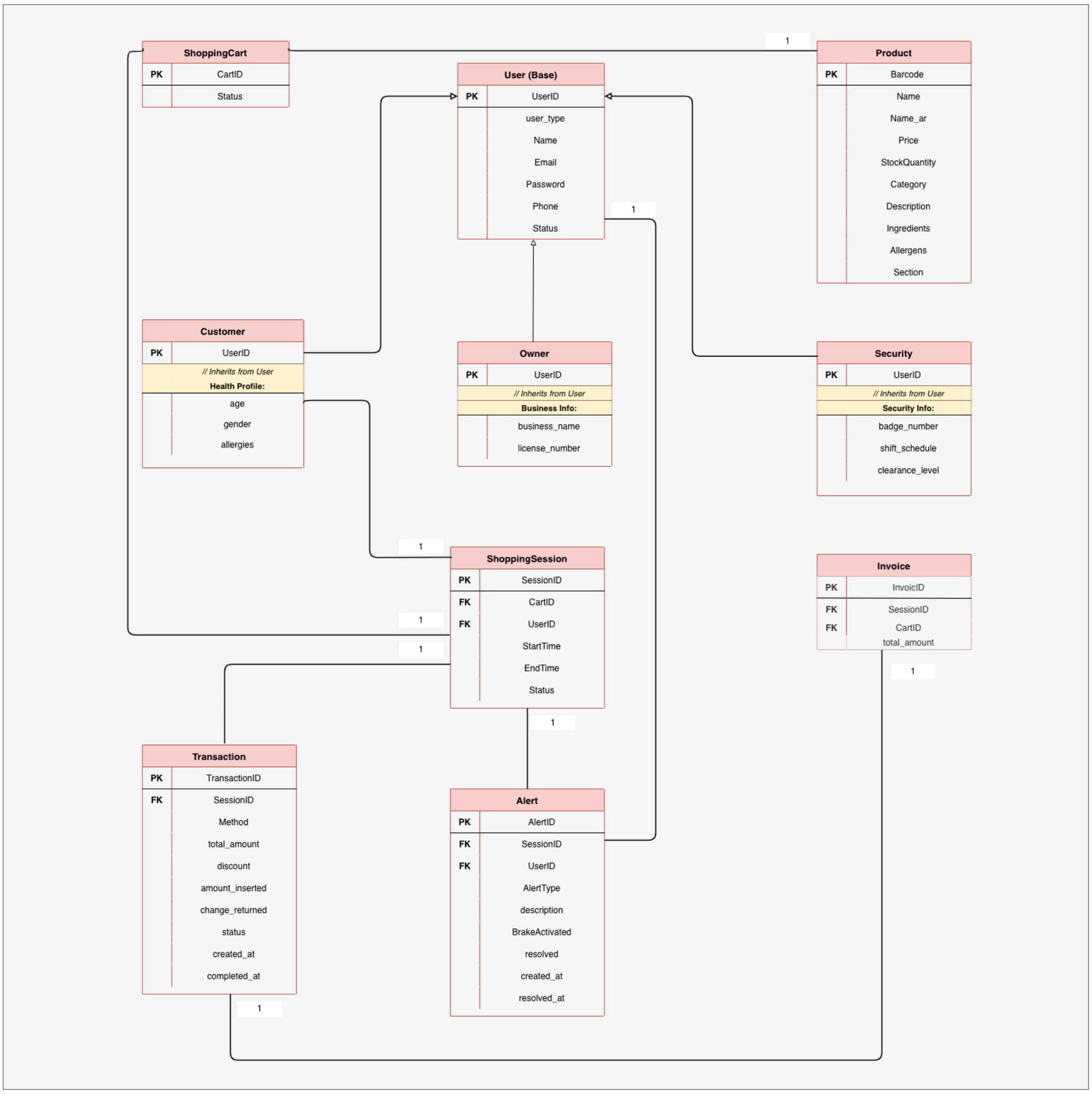


Figure 4.7: NEOSHOP ERD Overview

4.4 System Schematic

4.4.1 Cart System

The Cart System architecture integrates the Raspberry Pi 4 as the central processing unit, connected to the RFID module, OLED display, motor driver, and other peripherals within the smart cart. The system is responsible for product identification, real-time display, user interaction, and communication with backend services. Data flows between components through GPIO, SPI, and I2C interfaces, enabling efficient control and monitoring of cart operations. As shown in Figure 4.8, the schematic diagram illustrates all hardware connections and signal routing within the cart system.

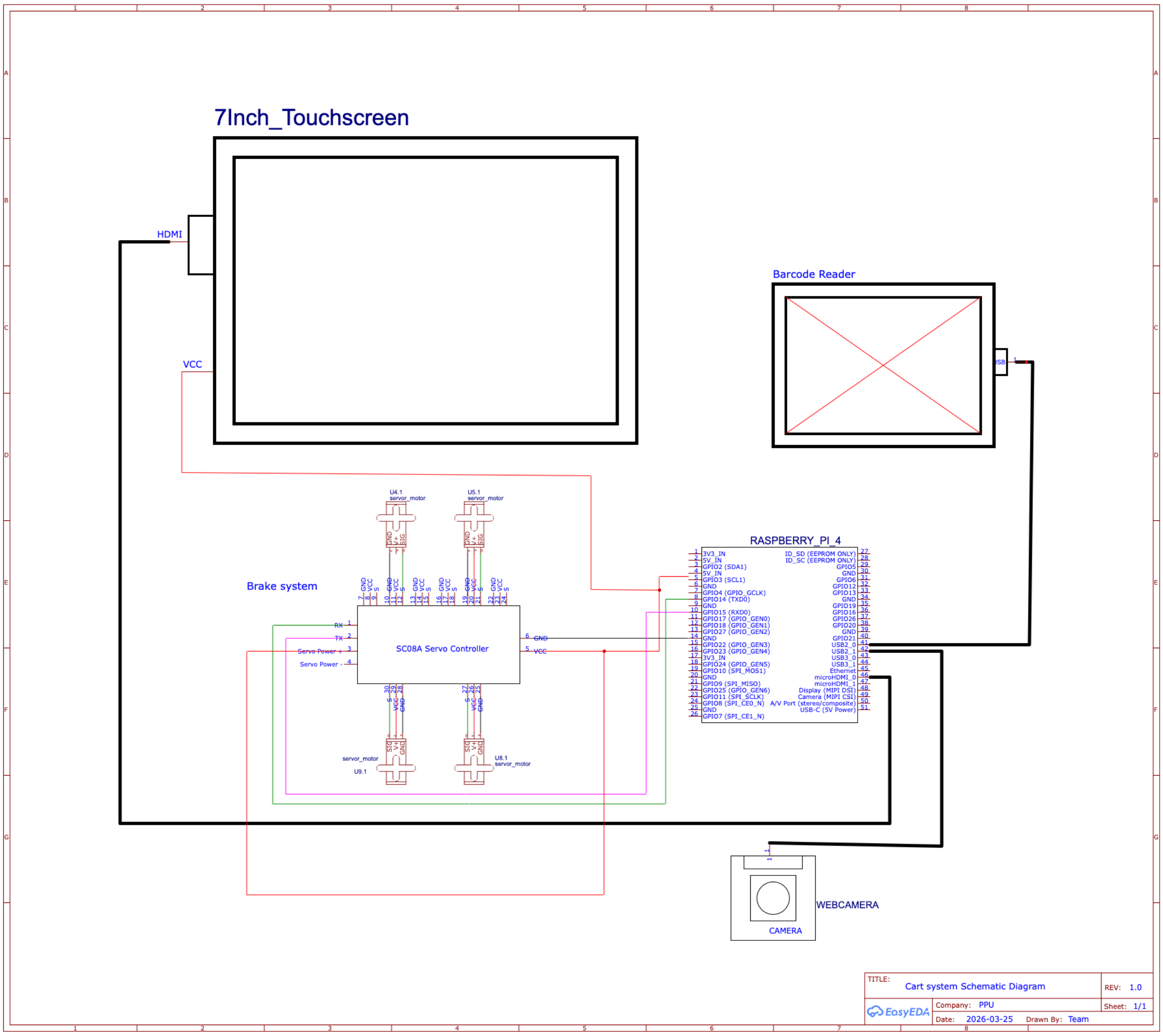


Figure 4.8: Schematic Diagram of the Cart System

4.4.2 Payment System

The payment system includes a coin acceptor (CH-926), a computer vision-based banknote authentication subsystem, an ESP32 controller, and associated interfaces.

Banknote authentication is performed by a composite system that detects the denomination, identifies counterfeit notes, and provides the total amount paid. Coins are processed via the CH-926 pulse output and interpreted by the ESP32. The system communicates transaction data securely with the backend using wireless protocols secured via WPA3 and TLS 1.3. As illustrated in Figure 4.9, the schematic diagram presents the complete wiring and component layout of the payment processing unit.

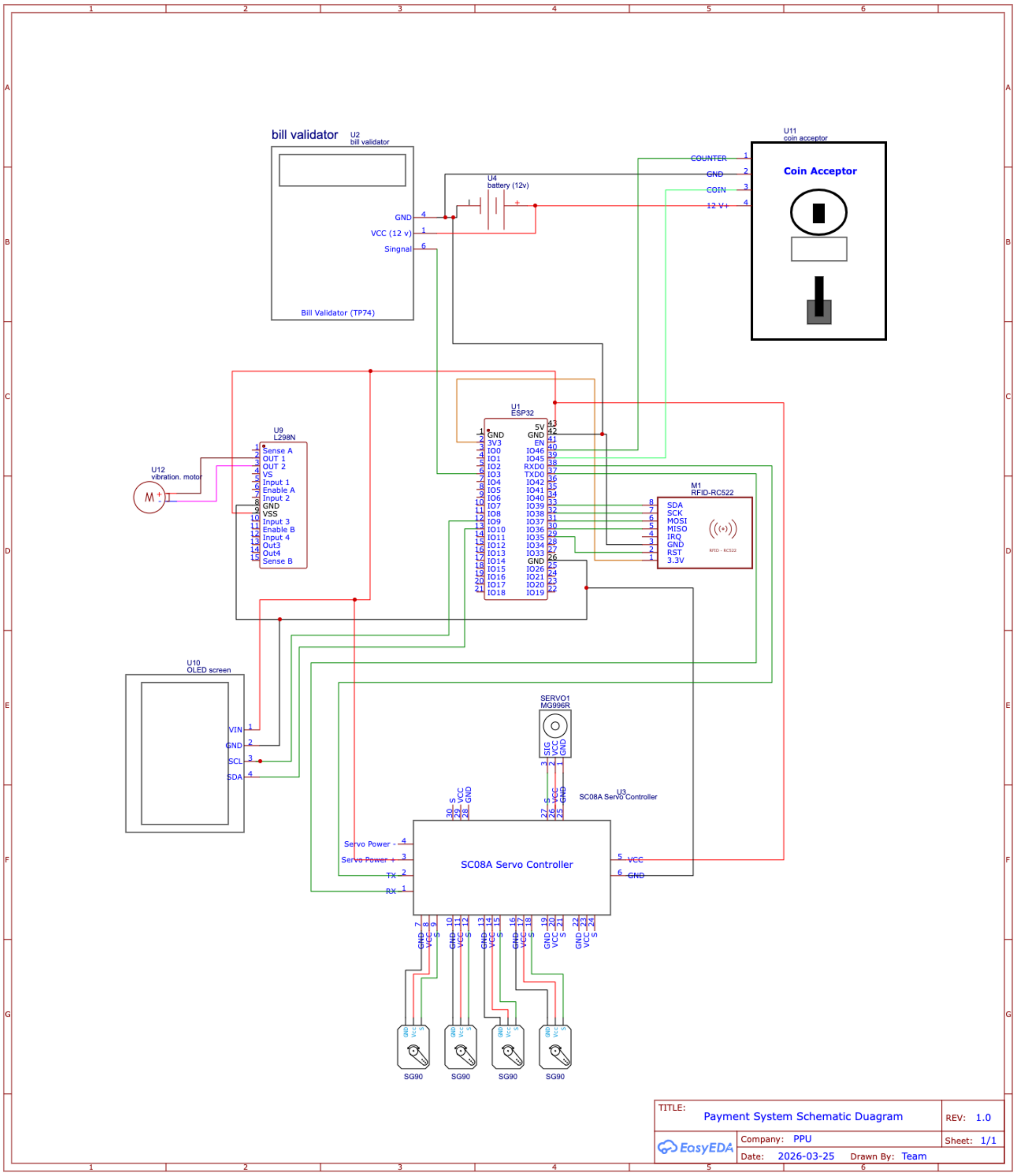


Figure 4.9: Schematic Diagram of the Payment System

4.5 Summary

This chapter presented the NEOSHOP system design through two main sections. The first section provided consolidated comparative tables for all hardware and software components, accompanied by well-defined justifications for each final selection based on critical criteria such as performance, cost efficiency, ease of integration, scalability, and overall system reliability. These comparisons ensured that each component was carefully evaluated and selected to meet the specific requirements of the proposed system.

The second section introduced six complementary diagrams BDD, IBD, Activity, Sequence, Parametric, and

Schematic which together offer a comprehensive representation of the system architecture from both structural and behavioral perspectives. These diagrams demonstrate how different components interact, how data flows within the system, and how various processes are executed in a coordinated manner.

Collectively, the presented design confirms that the proposed system satisfies all functional and non-functional requirements defined in Chapter 3, while also ensuring consistency, modularity, and future extensibility. As a result, this chapter establishes a solid, well-organized, and reliable foundation that supports the successful transition to the implementation phase. The use of a message broker further improves system reliability, scalability, and ensures smooth communication between subsystems.

Chapter 5

Conclusion

This project addresses the critical challenges faced in traditional retail environments, including long checkout queues, manual payment errors, undetected shoplifting, and the lack of actionable data for store owners. The proposed solution, NEOSHOP, is a fully integrated smart shopping cart and automated payment system that combines AI-powered computer vision, RFID-based checkout, and automated cash handling into a single cohesive platform. A Raspberry Pi 4 runs YOLOv8-nano for real-time object detection alongside ByteTrack for multi-object tracking, enabling continuous theft detection through hand movement monitoring, automatic cart braking, and real-time alerts to security staff and the owner dashboard. Product identification is handled via a laser barcode scanner, while RFID enables seamless checkout with invoice data delivered reliably to the payment unit through a message broker. The ESP32-based payment unit handles cash transactions with intelligent validation, greedy-algorithm change calculation, and automated dispensing, while a rule-based recommendation engine provides personalised, health-aware product suggestions without requiring historical user data. It is anticipated that NEOSHOP will reduce checkout time, minimise theft incidents, and provide store owners with actionable analytics, establishing a solid foundation for the implementation and evaluation phases of the project.

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نموذج إقرار وتوثيق استخدام أدوات الذكاء الاصطناعي في مشروع التخرج

اسم الطالب / أسماء الطلبة	أروى حجاحجة / بلال عدور / مؤمن بحيص
عنوان المشروع	NEOSHOP
الرقم / الأرقام الجامعية	221073 / 201089 / 231155
اسم المشرف الأكاديمي	وائل التكروري
الكلية / الدائرة	كلية تكنولوجيا المعلومات وهندسة الحاسوب
السنة / الفصل الدراسي	2026 الفصل الثاني

الغرض من الاستخدام	اسم الأداة	أماكن الاستخدام في التقرير (فصول / صفحات)	الأمر الأساسي المستخدم (Prompt)	هل تم تعديل الناتج؟	نسبة الاستخدام التقديرية	ملاحظات
توليد نصوص ☐	ChatGPT	صياغة الجمل في بعض الأماكن	اعد صياغة النص الآتي بلغة سليمة	نعم	10%	
☐ Gemini						
☐ Claude						
Microsoft Copilot ☐						
تدقيق لغوي ☐	ChatGPT	جميع فصول التقرير	تدقيق لغوي للنص	نعم	10%	
☐ Grammarly						
☐ Quillbot						
Word Editor AI ☐						
تلخيص محتوى ☐	ChatGPT	الملخص	لخصلي المشروع	نعم	2%	
☐ SMMRY						
☐ AI Notion						
☐ Scholarcy						
تحليل بيانات	Excel Copi-lot ☐	—	—	لا	0%	

AI Python As-sistants ☐						
☐ AI Tableau						
رسم مخططات / أشكال	☐ DALL·E	—	—	نعم	7%	
☐ AI Canva						
☐ AI Visio						
AI Lucidchart ☐						
كتابة أكواد برمجية	GitHub ☐ Copilot	—	—	لا	0%	
☐ AI Replit						
☐ Codeium						
توثيق مراجع	☐ EndNote	—	—	لا	0%	
☐ Mendeley						
☐ Zotero						
☐ ChatGPT						
أخرى (حدد) : ________						

إقرار فريق مشروع التخرج :

نقر بأن استخدام أدوات الذكاء الاصطناعي تم بشكل مسؤول وبما يتوافق مع السياسات الجامعية، وقد راجعنا وحررنا كافة المخرجات بما يعكس فهمنا الشخصي.

توقيع الطالب / الطلبة :
.... / .... / ....

التاريخ : ....